

Technical Implementation Proposal

Custom Enterprise Quran Learning Platform

Prepared for System Stakeholders & Investment Partners

1. Executive Summary & Vision

Our objective is to engineer a highly scalable, premium educational platform dedicated to Quranic studies. Standard learning management systems fail to deliver the rigorous audio quality required for *Tajweed* (pronunciation) and lack culturally aligned user experiences.

This proposal outlines a complete end-to-end technical strategy. We will begin with a highly efficient **Prototype Phase** utilizing an Enterprise-Grade Serverless Architecture. This approach minimizes upfront operational costs while validating the market. Crucially, because we are utilizing open-source core technologies (PostgreSQL, WebRTC), the entire application can be seamlessly transitioned to fully managed, globally distributed cloud environments when scaling to millions of users.

The "Future-Proof" Guarantee: The code written for this prototype will not be discarded. The frontend (React), database schema, and custom video engine are designed to easily decouple from our prototype cloud providers and deploy onto any major global cloud (AWS, Google Cloud, or local GCC infrastructure).

2. Platform Architecture Highlights

The platform will deliver a native, app-like experience directly within the web browser.

- **In-Platform Video Engine:** We will integrate a custom WebRTC engine (LiveKit). This guarantees sub-50ms latency and bypasses aggressive noise-cancellation codecs to preserve the raw audio fidelity necessary for accurate Tajweed correction. *No external Zoom or Google Meet links will be used.*

- **Automated Recording Pipeline:** Classes are automatically recorded via a server-side Egress worker. These recordings are securely processed and injected directly into the student's learning profile for revision.
- **Gamified Progression:** A sophisticated data schema tracks student XP, reading streaks (Mutaba'ah), and milestone progression to drive retention and motivation.
- **Asynchronous Materials:** Secure cloud storage integration allowing coaches to upload supplementary videos, audio clips, and PDFs.

3. Implementation Roadmap & Time Estimations (8-Week Delivery)

The project will be delivered in four Agile sprints across an 8-week timeline. Total estimated engineering effort is **125 – 145 Hours** across the full release cycle. You will have access to a live testing environment at the conclusion of each sprint.

Sprint 1: Foundation & Authentication

Calendar Duration: Weeks 1 – 2 | Estimated Effort: 30 – 35 Hours

Focus: Establishing the secure database and user management framework.

- Provision the serverless PostgreSQL database (Supabase).
- Implement secure Role-Based Authentication (Students, Teachers, Admins).
- Develop the foundational frontend UI (Next.js) with native Right-to-Left (RTL) Arabic support.
- Deploy the initial Teacher and Student profile views.

Sprint 2: Video Engine & Scheduling

Calendar Duration: Weeks 3 – 4 | Estimated Effort: 35 – 40 Hours

Focus: Delivering the core value proposition—live video classes and booking.

- Integrate the LiveKit WebRTC engine into the web interface.
- Configure custom, uncompressed audio profiles specifically for Tajweed.
- Build the interactive booking calendar and class scheduling system.
- Implement automated session notification pipelines.

Sprint 3: Recording & Cloud Materials

Calendar Duration: Weeks 5 – 6 | Estimated Effort: 30 – 35 Hours

Focus: Asynchronous learning and media retention.

- Configure LiveKit Egress to securely record sessions on the server.
- Build the pipeline to push .mp4 recordings into secure S3-compatible cloud storage.
- Link processed recordings automatically to the specific student/class database row.
- Enable teachers to upload and manage supplementary learning media.

Sprint 4: Gamification & Polish

Calendar Duration: Weeks 7 – 8 | Estimated Effort: 30 – 35 Hours

Focus: Student engagement, optimization, and Beta launch readiness.

- Develop the interactive Memorization (Hifz) tracking interface.
- Implement the gamification engine (calculating XP, badges, and streaks based on completed classes).
- Execute rigorous end-to-end testing across modern mobile and desktop browsers.
- Final deployment, DNS configuration, and handover for Beta Testing.

4. Cloud Architecture & Economics

To ensure capital efficiency, the prototype phase leverages managed serverless free-tiers. Below is the projected monthly cloud infrastructure cost during the testing phase.

Scenario A: Closed Alpha (Internal Testing, up to 20 users)

Infrastructure Component	Technology Provider	Estimated Monthly Cost
Frontend Hosting & Edge APIs	Vercel (Hobby Tier)	\$0.00
Database, Auth & Base Storage	Supabase (Free Tier)	\$0.00
Live Video Engine & Recording	LiveKit Cloud (Build Tier - 5k mins)	\$0.00
Total Estimated Alpha Cost:		\$0.00 / mo

Scenario B: Active Beta (Live Users, up to 100 students)

As real users are onboarded, upgrading the database and video engine ensures sufficient storage for HD video recordings and prevents service throttling.

Infrastructure Component	Technology Provider	Estimated Monthly Cost
Frontend Hosting & Edge APIs	Vercel (Hobby Tier)	\$0.00
Database & Media Storage (100GB)	Supabase (Pro Tier)	\$25.00
Live Video Engine (150,000 mins)	LiveKit Cloud (Ship Tier)	\$50.00
Total Estimated Beta Cost:		\$75.00 / mo

Scaling Note (V1 Production): Upon successful validation and scale, the infrastructure will transition to a global AWS architecture. For ~500 active students generating 360,000+ minutes of video monthly, managed global hosting will stabilize at approximately **\$350 - \$495 / month**, requiring zero rewrites of the core application.

Client / Investor Approval

Lead Cloud Engineering Team